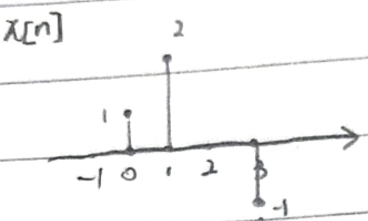
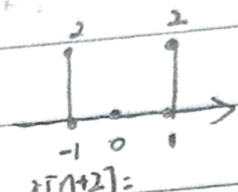
信号与系统：

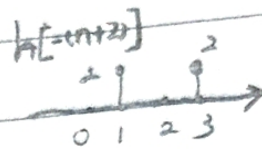
习题二

2.1

$x[n]=\begin{cases}1 & n=0\\ 2 & n=1\\ -1 & n=3\\ 0 & n=\text{其它}\end{cases}$ $h[n]=\begin{cases}2 & n=\pm1\\ 0 & n=\text{其它}\end{cases}$







(a.)

当 $n<-1$ 时，$y[n]=0$

$y[-1]=2\times1=2$

$y[0]=2\times2=4$

$y[1]=2\times1=2$

$y[2]=2\times2+(-1)\times2=2$

$y[3]=0$

$y[4]=-1\times2=-2$

当 $n>4$ 时 $y[n]=0$

(b.) 当 $n<-3$ 时，$y[n]=0$

$y[-3]=2$

$y[-2]=4$

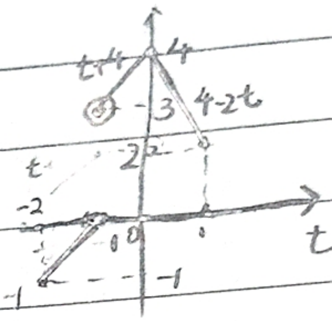
$y[-1]=2$

$y[0]=2$

$y[1]=0$

$y[2]=-2$

当 $n>2$ 时 $y[n]=0$

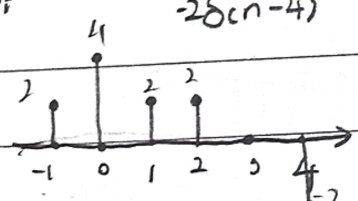


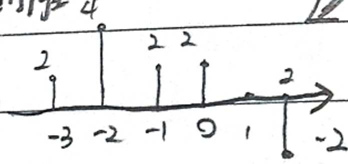
$\therefore y_1[n]=2\delta(n+1)+4\delta(n)+2\delta(n-1)+2\delta(n-2)-2\delta(n-4)$

$y_2[n]=y_1[n+2]$

$y_2[n]/y_1[n]$

$y_3[n]=y_2[n]$





设 $y(t)=h(t)*x(t)$

① $t<-2$ 时 $y(t)=0$

④ $1<t$ 时 $y(t)=0$

2.5: $y[4]=5 \Rightarrow N=4$

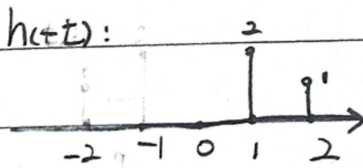
$\Rightarrow N>4$

$y[14]=0$

$\Rightarrow 14-N>9$

$\Rightarrow N<5$

2.8: $h(t)$:

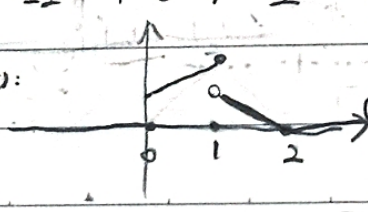


② $-2<t\leq-1$ 时

$y(t)=\delta(t+2)\cdot x(t+2)$

$=t+1$

$x(t)$:



③ $-1<t\leq0$ 时

$y(t)=\delta(t+2)\cdot x(t+2)+2\delta(t+1)\cdot x(t+1)$

$=2(t+1)+2-t=t+4$

④ $0<t\leq1$ 时

$y(t)=2\delta(t+1)\cdot x(t+1)=4-2t$

2.9. $h(t)=\begin{cases} e^{-2t} & t\geq 5 \\ 0 & 4\leq t<5 \\ e^{2t} & t<4 \end{cases}$

$\Rightarrow \begin{cases} t-\tau\geq 5 \\ t-\tau\leq 4 \end{cases}$

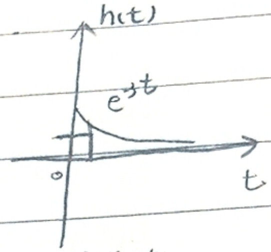
$\Rightarrow t-4<\tau\leq t-5$

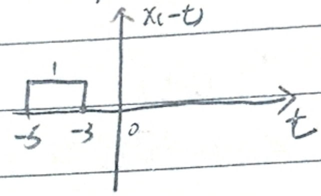
$\Rightarrow B=t-4 \quad A=t-5$

2.11: $x(t)=\begin{cases} 1 & 3<t<5 \\ 0 & 其它 \end{cases}$

$h(t)=\begin{cases} e^{-3t} & t\geq 0 \\ 0 & t<0 \end{cases}$

(a.) 记 $y(t)=x(t)*h(t)=h(t)*x(t)$





$t<3$时 $y(t)=0$

$3<t<5$时 $y(t)=\int_0^{t-3}e^{-3\tau}d\tau=-\frac{1}{3}e^{-3(t-3)}+\frac{1}{3}$

$5<t$时 $y(t)=\int_{t-5}^{t-3}e^{-3\tau}d\tau=-\frac{1}{3}e^{-3(t-3)}+\frac{1}{3}e^{-3(t-5)}$

(b.) 记 $k(t)=x'(t)=\delta(t-3)-\delta(t-5)$

$=\begin{cases} 1 & t=3 \\ -1 & t=5 \\ 0 & 其它 \end{cases}$

$g(t)=h(t)*g(t)$

$t<3$时 $g(t)=0$

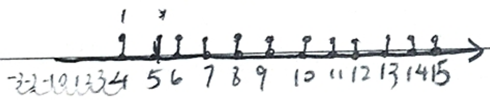
$3<t\leq 5$时, $g(t)=e^{-3(t-3)}$

$5<t$ 时 $g(t)=e^{-3(t-3)}-e^{-3(t-5)}$

(c.) $\frac{dy(t)}{dt}=g(t)$

2.4: $y[n]=x[n]*h[n]=h[n]*x[n]$

∴ $h[n]$



$x[k-n]=\begin{cases} 1 & -8+k\leq n\leq -3+k \\ 0 & 其它 \end{cases}$

当 $n<7$时 $y[n]=h[n]*x[n]=0$

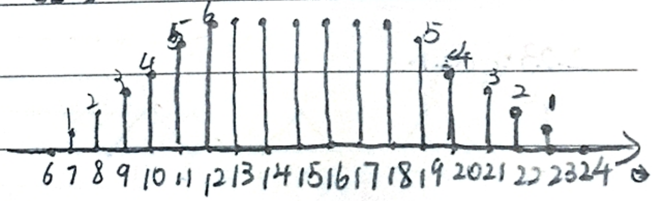
当 $7\leq n\leq 12$时 $y[n]=n-6$

当 $12<n\leq 18$时 $y[n]=6$

当 $18<n\leq 23$时 $y[n]=24-n$

当 $n>23$时, $y[n]=0$

∴ $y[n]$:



2.17:

(a) 由特征方程: $\lambda+4=0$

$\therefore \lambda=-4$

$\therefore y_h(t)=Ae^{-4t}$

设特解 $y_p(t)=Y\cdot e^{(-1+3j)t}$ $(t>0)$, $x(t)=e^{(-1+3j)t}$

$\therefore Y\cdot(-1+3j)\cdot e^{(-1+3j)t}+4Y\cdot e^{(-1+3j)}=e^{(-1+3j)t}$

$\Rightarrow Y(3+3j)=1$

$\therefore Y=\frac{1-j}{6}$

$\therefore y=A\cdot e^{-4t}+\frac{1-j}{6}\cdot e^{(-1+3j)t}$

又$\because$初始松弛: $y(0)=0$

$\Rightarrow 0=A+\frac{1-j}{6}$

$\therefore A=\frac{j-1}{6}$

又$\because$ $t<0$时, $y(t)=0$

$\therefore y=\left[\frac{j-1}{6}e^{-4t}+\frac{1-j}{6}e^{(-1+3j)t}\right]\cdot u(t)$

(b.) 记 $x_1(t)=e^{(-1+3j)t}$, $y_1(t)=\frac{j-1}{6}e^{-4t}+\frac{1-j}{6}e^{(-1+3j)t}$

$\therefore x(t)=Re\{x_1(t)\}$

由题意知

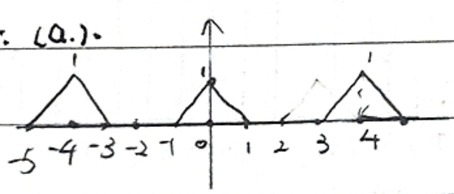
$$\frac{d}{dt}Re\{y_1(t)\}+Re\{y_1(t)\}=x(t)$$

$$\Rightarrow y(t)=Re\{y_1(t)\}$$

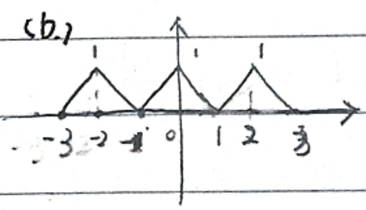
$$=Re\left\{\frac{j-1}{6}(e^{-4t}-e^{(-1+3j)t})\right\}$$

$$=-\frac{1}{6}\left[e^{-4t}-e^{(-1+3j)t}\right]$$

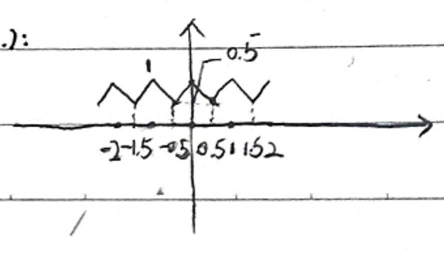
2.23: (a.)



(b.)



(c.):



(d.)